



Fitness Planner

Project Engineering

Year 4

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Bachelor of Engineering (Honours) in Software and

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Declaration

This project is presented in partial fulfilment of the requirements for the degree of Bachelor of Engineering (Honours) in Software and Electronic Engineering at Atlantic Technological University.

This project is my own work, except where otherwise accredited. Where the work of others has been used or incorporated during this project, this is acknowledged and referenced.

NOTE: The Project Title and Author Name appear in the header of each page. You must edit these "Document Properties" using the File > Info menu!

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1 Summary

This is a web-based application designed to help users organize and manage their weekly fitness routines in a simple and structured way. The goal of the project was to create an intuitive planner that allows users to build workouts and schedule them across different days of the week, improving consistency and planning in fitness activities.

The scope of the project includes creating, editing, and deleting workouts, as well as assigning them to specific dates within a weekly planner interface. Key features include a dynamic weekly schedule displayed in a column-based layout, a workout management system with detailed inputs (such as target area, reps, and equipment), and an interactive modal for assigning workouts to days. The interface was designed to be responsive and user-friendly.

The project was developed using a full-stack approach. The frontend was built with HTML, CSS, and JavaScript to handle layout, styling, and interactivity. The backend was implemented using Node.js and Express.js to manage API routes and handle data operations. Data was stored in JSON files, allowing persistence without the need for a full database. Communication between frontend and backend was achieved using the Fetch API.

The project successfully meets its goal by providing a functional and visually clear workout planner.

2 Poster



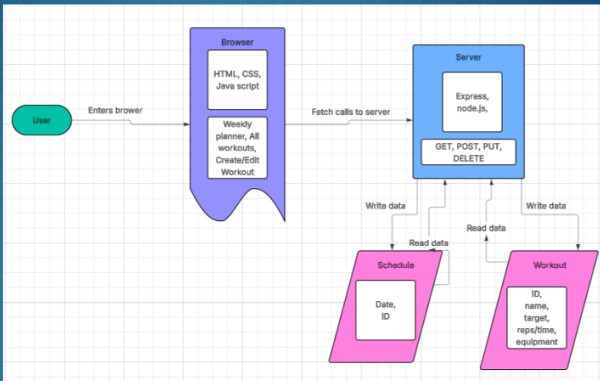
Atlantic Technological University

Fitness Planner
Chinonso Obasi
Beng (Hons) Software & Electronic Engineering

Summary

This project is a web-based workout planner that allows users to create, manage, and schedule workouts throughout the week. Users can add exercises with details like target area, reps, and equipment, then assign them to specific days using an interactive weekly planner. The app uses a Node.js and Express backend to store data in JSON files, ensuring workouts and schedules are saved.

Architecture diagram



```

graph LR
    User((User)) -- "Enters browser" --> Browser[Browser  
HTML, CSS, JavaScript  
Weekly planner, All workouts, Create/Edit Workout]
    Browser -- "Fetch calls to server" --> Server[Server  
Express, node.js  
GET, POST, PUT, DELETE]
    Server -- "Write data" --> Schedule[/Schedule  
Date, ID/]
    Server -- "Write data" --> Workout[/Workout  
ID, name, target, reps/time, equipment/]
    Schedule -- "Read data" --> Server
    Workout -- "Read data" --> Server
  
```

Technologies used

Software:
Node.js: Runtime environment that runs your backend server.
Express: Framework used to build the server and handle API routes.

Tools:
HTML: Used to structure the layout of the app.
CSS: Used to style and design the interface.
JavaScript: Handles frontend logic and interactivity.
Fetch API: Allows communication between frontend and backend.
Json: Stores workouts and schedule data locally.

Results

FITNESS PLANNER
workouts created: 4

Weekly Planner

Schedule workouts for each day of the week.

← →

Create Workout All Workouts

Monday 27 Apr	Tuesday 28 Apr	Wednesday 29 Apr	Thursday 30 Apr	Friday 1 May	Saturday 2 May	Sunday 3 May
sit ups 3x10 X bicep curls 3x10 X + Add	push ups 3x12 X sit ups 3x10 X plank 1min X + Add	+ Add	+ Add	+ Add	+ Add	+ Add

3 Introduction

Write a short introduction to the report.

Your introduction should state the goal of the project, and the motivation. Outline the scope of the project, i.e. the terms of reference.

You may add, as a final paragraph(s) a description of the layout of your report, however this is not absolutely necessary for short-ish report (< 50 pages) like this.

3 Background

3.1 Fitness Planning and Organisation

Maintaining a consistent fitness routine requires proper planning and organisation. Many individuals find it difficult to balance different types of workouts throughout the week, such as strength training, cardio, and recovery. Without a structured plan, workouts can become repetitive or inconsistent, reducing overall effectiveness. Traditional methods like notebooks or basic phone notes do not provide a clear weekly overview, making it harder to track progress or maintain balance.

A weekly planner system helps users visualise their schedule, distribute workouts evenly, and stay committed to their goals. By organising workouts by day, users can ensure they are targeting different muscle groups and allowing adequate recovery time.

3.2 Digital Fitness Applications

With advancements in technology, digital fitness applications have become increasingly popular. These applications provide tools for creating, editing, and managing workouts in a more efficient and interactive way. Many apps include features such as workout libraries, scheduling systems, and progress tracking.

A key advantage of digital planners is their flexibility. Users can easily modify workouts, rearrange schedules, and access their plans at any time. This improves usability compared to traditional methods and supports better long-term consistency.

3.3 Web Application Development

Modern web applications are typically built using a client-server architecture. The frontend is responsible for the user interface and experience, while the backend manages data storage and processing. This separation allows for more organised and scalable development.

Frontend technologies such as HTML, CSS, and JavaScript are used to create responsive and interactive interfaces. Backend technologies like Node.js and Express.js handle server operations and API requests. Data can be stored in various ways, and for smaller projects, JSON file storage provides a simple and effective solution without the need for a full database.

3.4 Relevance to This Project

This project applies the concepts of fitness planning and web development to create a functional workout planner. It combines a workout management system with a weekly scheduling interface, allowing users to organise their fitness routines in a clear and structured way. By using web technologies, the application demonstrates how a real-world problem can be solved through a simple, interactive digital solution.

4 Project Architecture

Your project architecture diagram should go here. This is an important section, and one most readers of your report will view.

Your diagram should be self-documenting. Use subsequent sections in your report to elaborate on technologies / software / hardware in your diagram.

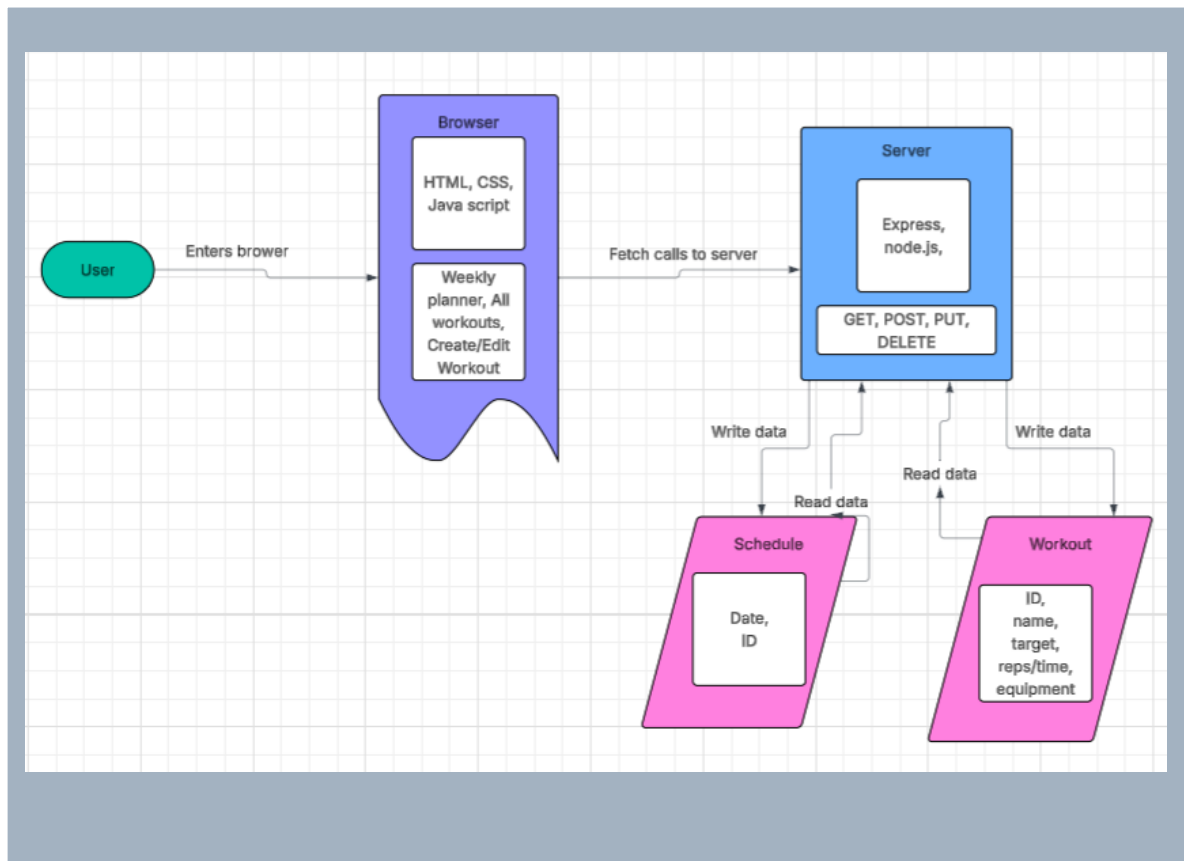


Figure 5-1 Architecture Diagram

5 Project Plan

5.1 Jira

The screenshot shows the Jira interface for a project named 'Final Year Project'. The 'Backlog' tab is selected, displaying a list of tasks organized into two sprints.

Sprints:

- FYP Sprint 1** (26 Nov – 10 Dec, 22 work items): 0 Done, 0 In Progress, 0 To Do.
- FYP Sprint 2** (28 Apr – 29 Apr, 24 work items): 0 Done, 0 In Progress, 0 To Do.

Task List:

Task ID	Description	Status	Assignee
FYP-24	For my fitness planner I want to remove the MySQL database and instead run a express server	DONE	-
FYP-25	For my fitness planner I will create data folder to store data for the workouts and the schedule.	DONE	-
FYP-26	For my fitness planner I want to create API endpoints in my server.js for workout and schedule.	DONE	-
FYP-27	For my fitness planner I want to give workouts random IDs so to not cause bugs when adding a...	DONE	-
FYP-28	For my fitness planner I will create a package.json file which will install and configure nodemon...	DONE	-
FYP-29	For my fitness planner I will create a fetch helper for API calls which makes all API calls go tho...	DONE	-
FYP-30	For my fitness planner I will fetch the workout and schedule data on startup.	DONE	-
FYP-31	For my fitness planner I will connect to backend workout endpoints.	DONE	-
FYP-32	For my fitness planner I will connect the planner to schedule endpoints.	DONE	-
FYP-33	For my fitness planner I will create a 7 day planer starting from monday	DONE	-

5.2 Github

The screenshot shows the GitHub repository page for 'Final Year Project'. The repository is public and has 1 branch (main) and 0 tags.

Recent Commits:

Commit Hash	Message	Time Ago	Commits
G00403720	Merge pull request #2 from G00403720/FYP-Completed	36cbb35 · 3 hours ago	7 Commits

Files:

File	Commit	Time Ago
data	Fitness-planner completed	3 hours ago
fitness-planner	Fitness planner	5 months ago
public	Fitness-planner completed	3 hours ago
Dockerfile	Fitness planner	5 months ago
docker-compose.yml	Fitness-planner completed	3 hours ago
package-lock.json	Fitness-planner completed	3 hours ago
package.json	Fitness-planner completed	3 hours ago
requirements.txt	Fitness planner	5 months ago
server.js	Fitness-planner completed	3 hours ago

Repository Statistics:

- Activity: 0 stars, 0 watching, 0 forks
- Releases: No releases published. [Create a new release](#)
- Packages: No packages published. [Publish your first package](#)
- Contributors: 1 (G00403720)

5.3 Notes on Content

On the backend, Node.js was used as the runtime environment to execute JavaScript outside the browser and run the server. Alongside this, Express.js was used as a web framework to simplify the creation of API routes, enabling the application to handle requests such as creating, updating, and deleting workouts, as well as managing the weekly schedule.

For development and testing, Visual Studio Code (VS Code) was used as the primary code editor. It provided useful features such as syntax highlighting, file management, and extensions to improve productivity. A web browser (such as Google Chrome) was used to run the application, test functionality, and debug issues using built-in developer tools.

On the frontend, HTML (HyperText Markup Language) was used to structure the layout of the application, including the planner, forms, and navigation. CSS (Cascading Style Sheets) was used to style the interface, controlling layout, colours, fonts, and responsiveness. JavaScript was used to implement interactivity and dynamic behaviour, such as rendering the planner, handling user input, and updating the interface in real time.

The Fetch API was used within JavaScript to enable communication between the frontend and backend, allowing data to be sent and retrieved asynchronously. Data was stored using JSON (JavaScript Object Notation) files, providing a simple and effective method of persistence without requiring a database.

5.4 Problem solving

The main problem I faced with this project was the API endpoints were unresponsive when I had it connected to a SQL database so I couldn't save the workout planner to it. I struggled with a way to fix this, so I turned to AI to give me an alternative. The alternative it gave me was using JSON data files which would save both the workouts created and the date the workout was added to the day.

This also allowed me to reuse the data stored in those files even when the server was turned off.

6 Ethics

User data privacy is a key concern. While the application stores relatively simple data (workout names, schedules, and preferences), it is still important to ensure that this data is handled responsibly. In this project, data is stored locally in JSON files rather than in a cloud database, reducing risks associated with data breaches. However, if the application were expanded, proper security measures such as encryption and authentication would be necessary.

Data integrity and reliability must be considered. Users rely on the planner to organise their workouts, so the system should consistently save and retrieve data correctly. Errors in scheduling or data loss could negatively affect the user experience or disrupt routines. The use of structured APIs and validation in the backend helps reduce these risks.

Another consideration is accessibility and usability. The interface should be clear, readable, and usable across different devices and for users with varying levels of technical ability. Thoughtful design choices—such as simple navigation, clear labels, and responsive layout—help make the application more inclusive.

While the app allows users to plan workouts, it does not provide professional medical or fitness advice. It is important that users understand the app is a planning tool only, and not a substitute for expert guidance.

7 Conclusion

The project allows users to create, edit, and delete workouts, as well as schedule them across a weekly planner interface. The application successfully integrates a front-end user interface with a back-end server, enabling persistent data storage and dynamic interaction.

The final product demonstrates a working system where users can organise their fitness routines in a clear and structured way. Key features such as a column-based weekly planner, interactive workout management, and modal-based assignment of workouts were implemented and tested. The interface is responsive and user-friendly, making it suitable for everyday use.

From a technical perspective, the project shows the successful use of core web technologies, including HTML, CSS, JavaScript, and Node.js with Express. It also highlights the ability to connect a front-end interface with a simple file-based data storage system using JSON.

8 References

[Course: 25-26: 6799 - - FULL STACK DEVELOPMENT | vlegalwaymayo.atu.ie](#)

[Course: 25-26: 3925 - - CLOUD](#)

[COMPUTING |](#)

[vlegalwaymayo.atu.ie](#)